

Noah Hickman

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EDUCATION

B.S. Mechanical Engineering - The University of Texas at Austin

May 2027

• **GPA:** 3.86; **Elective Coursework:** Powertrains/Vehicle Dynamics, Combustion Engine Processes, Robot Mechanism Design

SKILLS

Systems & Automation: Python, State Machine Logic, GIT, MATLAB, Root Cause Analysis, HIL/HITL Testing, P&ID

Simulation & Testing: ANSYS Mechanical (Finite Element Analysis), ATI Vision, Vector CANalyzer, LabVIEW, Simulink

CAD & Manufacturing: SolidWorks, 3D Printing, TIG Welding, GD&T, Lathe and Manual Mill, Soldering

General: Microsoft Office, Confluence, Jira, Agile, CompTIA IT Fundamentals+ (ITF+), trained in ITAR compliance

WORK EXPERIENCE

Vehicle Systems Engineering Intern - Relativity Space

May 2026 - Aug. 2026

- Developed automated control system logic and state machines for 6 high-pressure fluid systems for vehicle flight operations
- Led Thrust Vector Control actuator ground clearance testing automation development across 10 Aeon R and V engines
- Authored hardware-in-the-loop (HITL) validation and commissioning procedures to verify system operations against simulated sensor feedback and instrument states

Mechatronics Test Engineering Intern - Daimler Trucks North America

May 2025 - Aug. 2025

- Collaborated across Powertrain, ADAS, and Braking teams to analyze multi-system test failure modes and align automated reporting and data pipelines with live vehicle testing
- Deployed a Python test automation package to transcribe 320+ voice logs for fault event detection
- Executed over-the-road mechatronics validation tests with live DAQ, unifying 8,000+ files across 5 teams into SQL tables

Powertrain Integration Engineering Co-Op - Volvo Group

Jan. 2025 - May 2025

- Integrated fluids hardware, electronics, and sensors on HIL test benches to validate CARB-compliant engine packages
- Extracted and analyzed CAN/LIN bus data via ATI Vision and CANalyzer, resolving thermal and fluid anomalies across 56+ engine parameters (fuel mass flow, boost pressure, exhaust temp, etc.)
- Diagnosed powertrain HIL system faults through structured root cause analysis using MATLAB/Simulink documentation

TECHNICAL ACTIVITIES AND PROJECTS

Longhorn Racing Combustion (Formula SAE)

May 2025 - May 2026

Roles: **Engine Calibration Lead - Engineer**

- Assembled test engine with high-compression pistons and long-duration camshaft modifications to evaluate power gains
- Led dyno testing, engine calibration, and sensor instrumentation on a Honda CBR600RR powertrain across 4 test events, tuning for event-specific performance targets

Longhorn Racing Solar (Formula Sun Grand Prix, American Solar Challenge)

Aug. 2023 - May 2025

Roles: **Body Lead - Frame Sublead - Engineer**

- Led the 15-member Frame/Ergonomics subsystems and high-level design roadmap during a 1-year design cycle
- Automated chassis stiffness calculations via MATLAB, reducing ANSYS FEA simulation turnaround by 80%
- Utilized ANSYS and SolidWorks to validate and complete chassis design for legal road use based on ASC regulations
- TIG-welded 114-tube space frame using 3D-printed fixtures and custom-lathed pins to localize tabs to ± 0.001 -in tolerance

Flight-Weight Fluids Engineer - Texas Rocket Engineering Lab (Spaceport America)

Feb. 2024 - Jul. 2024

- Derived valve and fitting specifications from P&ID fluid schematics to satisfy high-pressure RCS fluid system requirements
- Anchored fluid system models by executing pressurant flow testing on an orifice flow bench via LabQuest DAQ, validating empirical discharge coefficients

Propulsion Engineer - Longhorn Rocketry Association (Spaceport America)

Aug. 2023 - Jul. 2024

- Sized a nozzle and impinging injector via NASA CEA and custom MATLAB mechanical design trade studies
- Executed material trade studies across 10 O-ring and lubricant materials, evaluating thermal and chemical compatibility

Hobbies: Project car building, record/vinyl collecting, running/lifting, and solo traveling; Member of UT ASME